JLU-SPH | BYEGERM

Community Collaborative Emergency Response Guide

Respiratory health in nursing homes, kindergartens and primary care settings



Connect an early signal to a clear next action

A community testing result becomes useful when someone receives it, understands its limits and takes responsibility for the next step. This guide connects community observation, preliminary screening, clinical assessment and public health follow-up.

Who should use this guide

The intended readers are nursing home and kindergarten managers, designated health staff, primary healthcare teams, receiving hospitals and laboratories, and public health coordinators. Project teams can also use it to test whether their device and reporting design fit an actual response pathway.

What comes from Stage 2

Stage 2 combines public health consultations on surveillance with clinical interviews on testing and referral. It positions ByeGerm as a proposed rapid, multiplex, portable and deployable preliminary screening node connected to professional assessment and confirmation.

Status and operating boundary

This is a proposed collaboration framework. It does not establish regulatory authorization, clinical validation or approval to deploy ByeGerm. Research and demonstrations should use simulated cases and non-clinical materials under institutional arrangements. A research prototype must not guide patient care.

Any future service needs authorization, validated instructions and review by clinical, public health, ethics and data-governance leads. Clinical assessment and official reporting obligations take precedence. Enter local definitions, deadlines and contacts in Appendix A.

Find the task you need

Task	Page
Assign accountable people and backups	3
Respond when symptoms or a concerning pattern appear	4
Check screening readiness and interpret result categories	5–6
Refer a person or escalate a possible cluster	7
Adapt the workflow and protect participants	8–9
Prepare, evaluate and improve the service	10
Complete local settings and practical records	11–13
Review the source basis and reference boundaries	14

Give each handover a named owner

Before any exercise or authorized service begins, name one incident coordinator for each site and one backup. A person may hold more than one role only when competence, workload and access controls allow it.

Role	Accountability and handover
Site coordinator	Activate the site procedure; arrange support, contacts and staffing; track unresolved actions; confirm that the receiving service has accepted each handover.
Caregiver or teacher	Notice and record changes; keep the person supported; contact designated health staff and the family through the agreed route. Do not diagnose or select treatment.
Authorized health professional	Assess symptoms and urgency; decide on examination, referral and any interim precautions; give follow-up and safety-net advice.
Authorized test operator	Work only within training and approved instructions; verify identity and sample linkage; check quality controls; record limitations; report invalid runs and faults.
Receiving hospital or laboratory	Accept the clinical or specimen handover; select appropriate testing; communicate confirmation, interpretation and further management to the responsible clinician.
Public health or disease control contact	Assess reported patterns; define investigation and reporting needs; advise on outbreak response and criteria for de-escalation.
Data custodian	Control access, secure records and transfers, maintain the identity-code key separately, and implement the locally approved retention and incident procedure.
ByeGerm project team	Provide technical documentation, training materials and research feedback channels. Refer clinical and public health decisions to the authorized services.

Close every handover

The sender records the recipient, time, information supplied and requested action. The receiver confirms receipt and ownership. If acknowledgement does not arrive within the locally agreed interval, the sender calls the backup contact and records the escalation. Sending a message alone does not complete a referral.

Keep authority clear

A site can initiate its immediate care and contact procedure without waiting for a test. Formal outbreak declarations, compulsory restrictions, prescribing and clearance for return belong to the competent authorities or clinicians under local rules.

Act on the person before the test

Use this sequence when a resident, child or staff member develops concerning respiratory symptoms, or when staff notice an unusual pattern of similar illness. Adapt it with the local clinical lead before use.

1 Recognize and document the change

Record when symptoms began or were first noticed, the setting and any relevant changes in attendance or illness among linked people. Record observations and the person’s account without assigning a pathogen or diagnosis.

2 Seek urgent help when needed

Breathing difficulty, reduced responsiveness, or a child who cannot feed or drink require immediate professional assessment. These examples are not a complete triage rule. Activate the local emergency pathway whenever serious illness is suspected; do not wait for screening, internet access or a referral form. [2, 3]

3 Provide supported separation and basic precautions

Where feasible, move a symptomatic person away from crowded activities while keeping appropriate supervision and essential care. Use hand hygiene, respiratory etiquette, ventilation and protection specified by local infection-prevention procedures. Keep a child with a suitable supervising adult; avoid leaving a dependent resident alone. [2, 3]

4 Contact the responsible health professional

Give the symptom and exposure summary, the person’s vulnerabilities and the help already arranged. The health professional decides whether immediate referral, examination, specimen submission, observation or eligible screening is appropriate. Notify the parent, guardian or chosen contact as appropriate to capacity, consent and local rules.

5 Screen only within an authorized pathway

If screening is authorized and clinically appropriate, use the eligibility and quality checks on page 5. If no authorized test or trained operator is available, continue the clinical and public health pathway without it. Record why the test was not performed.

6 Assign the next action and confirm follow-up

Apply the result pathway on page 6, confirm the receiver on page 7 and set a named follow-up owner and review time. Escalate a concerning linked pattern in parallel, even if individual tests are negative or unavailable. Use Appendix B for the person and Appendix C for the event.

Make a result traceable and interpretable

This guide describes collaboration around a test. Sampling technique, specimen type, test timing, reaction conditions and disposal procedures must come from the authorized device instructions and approved local protocols.

The proposed five-target panel

Target in Stage 2	Code	Interpretation boundary
Respiratory syncytial virus	RSV	A target-specific result; other causes of illness remain possible.
Human metapneumovirus subtype B	hMPV-B	A subtype B panel cannot be described as covering all hMPV.
Streptococcus pneumoniae	SPN	Detection alone does not establish the cause of symptoms.
Bordetella pertussis	BP	Follow the clinician and public health pathway for assessment.
Haemophilus influenzae	HI	Do not infer strain, type or invasive disease from this label.

These are the design targets reported in the manuscript, not a claim that the panel has been clinically validated. In particular, a bacterial nucleic acid signal may require interpretation in relation to specimen type, carriage and the clinical picture. A healthcare professional must interpret any future result.

Before a run

- Confirm the permitted use, trained operator, eligible person and appropriate consent process. Check the instructions for specimen suitability and timing.
- Use a coded record and verify the person-to-sample link. Check device status, consumable identity, expiry, storage and required quality controls.
- Confirm that a clinician can receive abnormal results, that confirmation is accessible, and that secure recording and downtime arrangements work.

During and after a run

Record the device and software version, operator, sample and run identifiers, consumable lot, collection time, run time and control status. Keep raw output where the approved protocol requires it. Preserve the original result and any later correction with an audit trail.

Treat failed controls, unacceptable specimens, unresolved contamination concerns or device faults as unusable results. Follow the approved repeat or alternative-testing procedure; do not convert a fault into a negative result. Stop affected testing when safe operation or result reliability is uncertain and notify the technical and clinical leads.

Decide by urgency and context

The following pathways apply only to an authorized future screening service. A result informs professional assessment; it does not independently authorize treatment, release from precautions or return to group activities.

Finding	Required next action	Accountable receiver
Serious illness suspected Any test result	Activate emergency assessment immediately. Send the symptom summary and available results when safe to do so.	Emergency service or receiving clinician
Presumptive positive One or more targets	Notify the responsible clinician, explain that the result is preliminary and arrange confirmation or further assessment as directed. Follow applicable public health reporting routes.	Clinical lead; public health contact when indicated
Negative result Symptoms persist or concern remains	Explain the limited panel and the possibility of an undetected infection. The clinician decides on further testing, observation or referral. Escalate deterioration.	Responsible clinician and follow-up owner
Negative result No current clinical concern	Document the advice and follow-up plan. Continue routine precautions. Do not use the result as a general certificate of non-infectiousness.	Designated health professional or care lead
Invalid or inconclusive result	Record that no usable conclusion is available. Seek repeat or alternative testing under the approved protocol; assess clinical urgency independently.	Authorized operator and clinical lead
Possible linked cluster Any or no test results	Contact the designated public health service through the local pathway. Share the minimum event information and continue individual care.	Site coordinator and public health contact

Interpret uncertainty explicitly

A negative result does not exclude pathogens outside the panel, a specimen taken at an unsuitable time, inadequate collection or infection below the validated detection limit. A positive signal is not, by itself, a measure of severity or infectiousness. Multiple detected targets do not automatically establish clinically significant co-infection.

Do not make autonomous treatment decisions

Medication, including antibiotics, requires professional assessment. Use the diagnosis, confirmatory evidence and clinical context to decide treatment. Repeated community testing must not delay assessment of a person who remains unwell.

Send the information the receiver needs

Manage individual care and a possible community event in parallel. A suspected cluster can require public health attention before any pathogen has been confirmed.

When to escalate a pattern

Contact the local public health service when linked respiratory illness appears unusual for the site, illness becomes unexpectedly severe, attendance or staffing changes suggest spread, or the local reporting trigger is reached. Local case definitions and reporting duties govern formal notification. Escalate concerns even when a formal reporting threshold has not yet been confirmed.

Individual clinical handover

- Identify the sender, callback route, receiving service and urgency. Use the minimum identifiers required for safe care and provide direct contact details securely.
- Summarize symptom onset, current concern, relevant vulnerabilities and the clinician’s assessment if available. Describe actions already taken and support needed.
- State whether a test was performed, the exact targets and result status, collection and result times, quality-control status and known limitations. Include confirmation status.
- Request a specific action, such as urgent assessment, confirmatory testing or advice on observation. Record acceptance, transport arrangements and follow-up ownership.

Event notification to public health

Provide the site and event code, first onset date, number of affected people and population at risk, relevant rooms or classes, severity, possible links, and the number tested with results by category. Distinguish suspected, preliminarily detected and clinically confirmed cases. Explain whether counts represent people, samples or test runs; do not mix them.

Add measures already implemented, unresolved needs and a secure callback route. Keep an event line list only within the authorized system. Share identifiable information only when necessary and through the required reporting or care channel. Appendix C supports the initial situation report and subsequent updates.

If the usual route fails

For urgent illness, use the emergency route immediately. For other escalation, try the designated backup or direct telephone route, log unsuccessful attempts and keep essential care running. During connectivity loss, use protected paper records and reconcile them with the authorized system once service returns. Record entry time separately from the original event time.

Confirmation and feedback

The receiving clinician or laboratory selects the appropriate confirmatory method and specimen route. Specimen collection, packaging and transport must be arranged by authorized personnel under applicable protocols. The sender tracks whether confirmation and management advice return; any material correction must reach earlier recipients.

Keep care practical in each setting

The collaboration roles remain the same, but supervision, communication and access to healthcare differ. Agree these adaptations with the people who work in each site.

Nursing homes

Designate a staff member who knows the resident's usual function and communication needs. Record changes from that baseline and seek clinical assessment promptly when staff or the resident are concerned. Include relevant underlying conditions in the secure clinical handover. Arrange assistance for hearing, vision, mobility or cognitive needs without assuming that age removes decision-making capacity.

Plan how essential medication, meals, hydration, personal care and family contact continue during temporary separation. Any movement restriction should follow the competent clinical or public health advice and be reviewed for necessity and duration. A site coordinator should track connected rooms, staff assignments and transport requirements without circulating resident identities broadly.

Kindergartens and childcare settings

Name the responsible staff member, the guardian contact and the healthcare referral route. Keep an unwell child supervised in a suitable space and contact the guardian promptly. Explain any proposed procedure in age-appropriate language, seek guardian authorization as required and respect the child's expressed discomfort. A teacher must not be expected to perform clinical sampling merely because a device is portable.

Record room or class links and attendance changes to support public health assessment. Give families a clear explanation of what is known, what remains uncertain, the next action and how updates will arrive. Use local health and education requirements for attendance, temporary measures and return; a single negative screen does not settle those decisions.

Primary care and remote settings

Confirm a receiving clinician and laboratory before planning decentralized screening. Test the telephone route, power supply, stock conditions and feasible specimen or patient transport. Keep an offline record option and make the fallback pathway usable when the device, network or consumables are unavailable.

Adapt information to local language, literacy and accessibility needs. If trained personnel, validated operating conditions, secure handling or timely referral cannot be provided, pause the screening component and use the established healthcare route. Portability does not remove these requirements.

Apply an equitable response

Screening availability, ability to pay, disability, language or willingness to join research must not determine access to essential care. Record practical barriers and raise them with the coordinator so that referral and communication can be adjusted.

Protect the person behind the record

The manuscript's C.A.R.E.S. framework provides the ethical basis: Compassion and Humanistic Care, Autonomy and Informed Consent, Reliability and Transparency, Equity and Accessibility, and Safety.

Explain choices before participation

Explain the purpose, procedure, likely burden, limitations, result recipients and intended data use in accessible language. Separate research participation from care. A refusal to join research must not reduce access to routine services. For a person who can decide, obtain their own informed agreement; a care worker or relative does not automatically have authority to consent on their behalf.

For children or adults who lack the relevant capacity, follow the applicable guardian or legally authorized representative process, seek the person's participation or assent where appropriate, and document the basis of authorization. Emergency care follows the applicable emergency procedure. Do not describe every public health reporting activity as optional consent; establish its lawful basis with the responsible institution.

Use the minimum necessary information

Use coded case and event records for coordination wherever possible. Keep the identity-code link separately with restricted access. Limit identifiable clinical information to those who need it for care or authorized reporting. Coded data can still identify someone when combined with other information and must be protected.

Use institution-approved access controls, secure transfer and an agreed retention and deletion schedule. Configure storage location and any cross-border transfer only after the responsible institution has reviewed the applicable requirements. This guide does not certify any particular server, encryption system or legal compliance status.

Use clear result language

Preliminary detection: "The screen detected a signal for [target]. This is a preliminary finding. The healthcare team will assess it and advise whether confirmation or further care is needed."

Negative screen: "No target on this panel was detected in this sample. The panel cannot test for every cause of illness. Please follow the care plan and seek help if symptoms worsen."

Invalid result: "This run did not give a usable result. The healthcare team will advise on repeat or alternative testing. Care decisions will continue based on symptoms."

Communicate a possible cluster without stigma

Use a designated spokesperson. Share practical actions and the next update time; distinguish concern from confirmation. Do not name affected residents, children or classes publicly when this could identify individuals. For the Wiki, use simulated examples and aggregated, disclosure-checked summaries; do not upload case forms, identifiable photographs or detailed line lists.

Rehearse the pathway before relying on it

Preparation is complete only when the site can move from an initial concern to an accepted referral, documented advice and follow-up, including when the test is unavailable.

Agree the service before activation

Complete Appendix A with the responsible leads. Confirm permitted use, trained operators, eligibility, quality controls, supplies, safe handling, infection prevention, data governance and referral access. Document who can start, pause and resume the service; resolve missing prerequisites before involving participants.

Run scenario exercises

- A resident has concerning breathing symptoms before a run finishes. Staff must activate emergency assessment without waiting for a result.
- A child remains unwell after a negative screen. Staff must arrange clinical advice, explain the panel limits and confirm guardian follow-up.
- Several linked people become ill while results are negative or unavailable. The coordinator must reach public health and report an uncertain pattern accurately.
- Controls fail, power is lost or the network stops. Staff must pause affected testing, keep records protected and use the fallback care route.

Measure the process without claiming clinical benefit

Measure	Definition to record
Time to professional contact	Elapsed time from the first recorded concern to successful clinician or emergency-service contact; report missing times separately.
Handover completion	Accepted referrals divided by referrals initiated, with the observation period and unresolved cases stated.
Result quality	Invalid runs divided by all runs, with repeat runs identified separately from people tested.
Follow-up completion	Records with documented review or outcome divided by records requiring follow-up in the agreed period.
Accessibility and burden	Record communication barriers, unsuccessful referrals, reasons for non-participation and feedback from participants and staff.

Set operational targets locally and label them as targets. Claims about diagnostic accuracy, reduced transmission or improved outcomes require an appropriate evaluation design and evidence; process metrics alone cannot establish them.

Review and close the loop

Use Context, Dialogue, Reflection and Iteration to review the event, missing perspectives and required changes. Assign each change an owner and deadline. Before closure, address outstanding care, reporting, record reconciliation and authorized de-escalation. Record the next review date and version.

Complete before operational use

Use this template with the responsible site, clinical, public health and data-governance leads. Blank fields mean the corresponding arrangement has not been documented. This is a planning record, not an authorization certificate.

Local role or route	Named person or service	Secure contact and backup
Site and incident coordinator		
Responsible clinical service		
Emergency assessment route		
Receiving hospital and laboratory		
Public health reporting contact		
Data custodian and technical support		

Site name and scope of service

Permitted device use and instructions version or simulation-only status

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Operator competence and eligibility criteria reference

Local reporting triggers and deadlines with source and effective date

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Handover acknowledgement interval and escalation route

Confirmation and transport arrangements including fallback

Consent process data access storage retention and incident procedure

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Clinical and public health review names dates and next review date

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Before activation: verify supplies, operating conditions, quality controls, infection-prevention arrangements, secure records and downtime procedures. Record unresolved items and keep the affected activity paused until the designated lead resolves them.

Screening and clinical handover

Use only within an authorized service or label clearly as a simulated exercise. Keep identifiable contact details in the protected care record; use the case code on routine coordination copies.

Record status □ Simulation □ Authorized service Case code _____

Site / room code _____ Date and recorder _____

Symptoms onset and first observation time

Current concern relevant vulnerabilities and professional assessment

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Urgent help requested actions taken and times

Consent or other applicable authority and communication assistance

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Screening field	Entry
No test performed or reason if not eligible	
Device / software / operator / run ID	
Specimen code / type / collection time	
Consumable lot / expiry / control status	
Target-specific output / result time / limitations	
Repeat or confirmation requested and owner	

Recipient service contact time acknowledgement and requested action

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Advice provided to participant or guardian and by whom

Follow-up owner review time and deterioration contact route

Confirmation outcome corrections and final handover status

Retain the original result. Date and attribute any correction, and inform earlier recipients when the correction changes their decisions. Handle, store and dispose of this record under the local approved procedure.

Situation report and review log

For a possible linked event, record what is known and what remains uncertain. This template supports coordination and does not replace a mandatory reporting form or official surveillance system.

Event code _____ Site _____ Update no. _____

Reporting period _____ Prepared by / callback _____

Situation field	Entry with time and source
First onset / affected rooms or groups / possible links	
Population at risk / symptomatic people / severe cases	
People screened / runs performed / invalid runs	
Preliminary results by target / confirmation status	
Public health contact / time / acknowledgement	
Actions advised / actions implemented / remaining needs	

Count people, specimens and runs separately. Define the case categories and time window with public health. Keep the coded line list in the authorized system; do not attach personal information to public updates.

Track actions until a receiver accepts them

Action or decision	Owner	Due or review time	Status or acceptance

Next update time recipient and channel

Closure basis competent advice received unresolved follow-up and owner

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After action review

Context and Dialogue What happened and whose experience was heard

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Reflection and Iteration What changes who owns it and by when

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Version used _____ Review date _____

SOURCE BASIS

How the manuscript became this guide

This is a source-based project guide with proposed operational detail. It is not an official clinical guideline, a regulatory submission or a record of expert approval of this document.

1 Primary project source

ByeGerm Human Practices English manuscript, supplied DOCX. Principal basis: Stage 2 DEFINE, including the public health and disease control roundtable, the interview with Deputy Dean Changgui Kou, and interviews in respiratory medicine, laboratory medicine, infectious diseases, pediatrics and geriatric medicine. Supporting basis: the Reflection Cycle and C.A.R.E.S. Ethical Evaluation Framework.

Source contribution	Translation into this guide
Community Collaborative Emergency Response Guide proposed in Stage 2	Community observation, screening, professional assessment and public health follow-up are joined into one accountable pathway.
Rapid multiplex portable and deployable design goals	Readiness, quality-control, referral and downtime requirements are made explicit; no numerical performance claims are added.
Five-target selection and clinical interviews	Exact target names are retained and result interpretation is separated from diagnosis, confirmation and treatment.
C.A.R.E.S. and the four-step reflection cycle	Consent, accessibility, privacy, safety and structured review are carried into the proposed workflow.

Role assignments, the six-step response sequence, handover rules, templates and process measures are drafting recommendations developed for this guide. They are not presented as actions already implemented, quotations from interviewees or commitments made by partner institutions.

2 External public health background

World Health Organization. Pneumonia in children. Background on urgent signs in children, prevention and the need for clinical care. Accessed 16 October 2026.

WHO fact sheet on pneumonia

3 World Health Organization. Human metapneumovirus infection. Background on respiratory symptoms, severe illness, vulnerable groups and basic prevention. Published 10 January 2025; accessed 16 October 2026.

WHO questions and answers on hMPV

Local review and version control

External background supports general health information; it does not validate ByeGerm. Before operational use, the responsible institution must attach current national and local rules, device instructions, clinical escalation criteria and approved data procedures. Review this document when those sources, the panel or the workflow change.

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